# Systems Engineering - Midterm Review

*Team 9 - Semantic Transmitter - "Semantic Audio Communication using GenAI"*

André Jacinto, João Abade, Carlos Sousa, José Santos, Eduardo Medeiros, Nuno Vieira, Guilherme Ferreira, Rodrigo Resende, Inês Faleiro, Teresinha Teixeira, Tomás Sousa

**Faculdade de Engenharia - Universidade do Porto**

April 21, 2026

1. Introduction
   - Team Organization
   - Challenge Description
2. Validated Requirements
3. Outcome of the Market Survey
4. System Breakdown Structure
5. Functional Architecture
6. Measures of Effectiveness and Performance
7. System Concept
8. Work Breakdown Structure + Gantt
9. Risk Management
10. Conclusion
    - Internal Evaluation
    - Next Steps

**Presenters**:

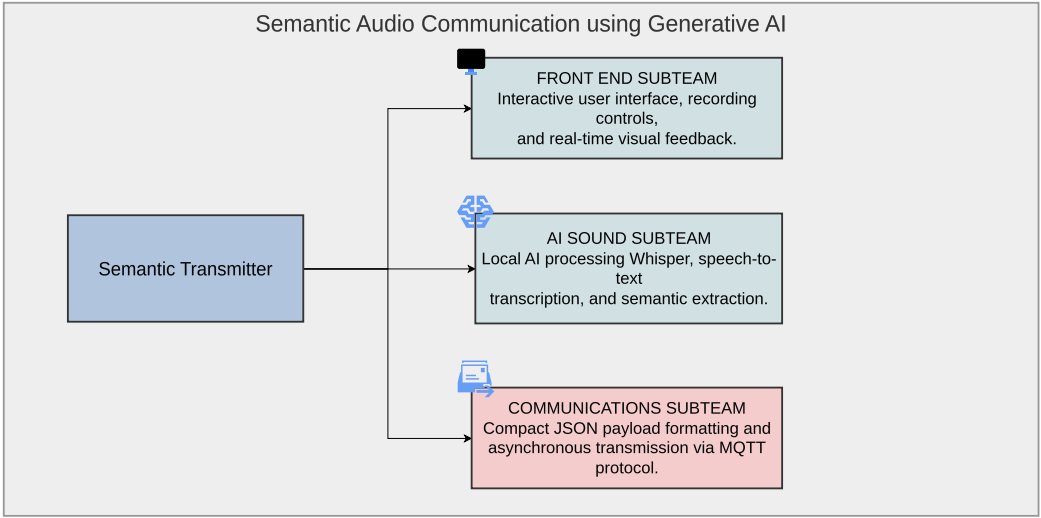
- João Abade
- Guilherme Ferreira
- Nuno Vieira
- José Santos
- Eduardo Medeiros

**Supervisor:**

- Professor Rui Lopes Campos

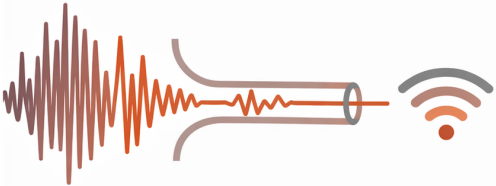
U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

# Team organization



# Challenge Description

## The Problem (Status Quo)



### Header: Traditional Voice Messaging

* High bandwidth consumption.
* Severe delivery delays on unstable links.

## The Challenge/Our Solution

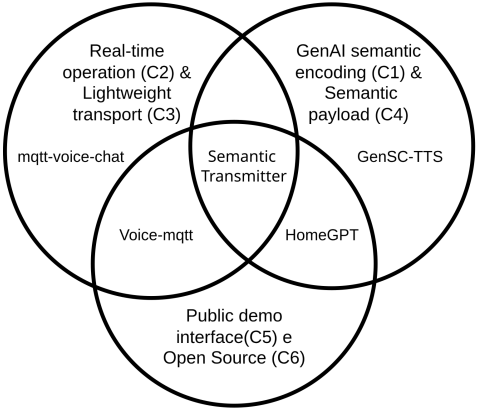
### Header: Semantic Transmission

* Edge AI local meaning extraction.
* Smaller payload containing semantic information.
* Capable of working in unstable network conditions.

# Validated Requirements

| Category | Validated requirements |
|---|---|
| **Communication** | Asynchronous message sending |
| | Standardized semantic payload with transcript and metadata |
| | Delivery-state feedback |
| | Measurable bandwidth reduction vs. raw audio transmission |
| **User Interface** | Explicit start/stop recording controls |
| | Dynamic transcript display after processing |
| | Real-time status feedback |
| | Graceful, non-disruptive error handling during demos |
| **AI STT Model** | Model runs locally; retry path for capture/transcription failures; transcript shown before transmission; raw audio discarded after transcription; low end-to-end latency |
| **Validation concept** | Peer Reviews & Checklist Comparison |

# Outcome of the Market Survey



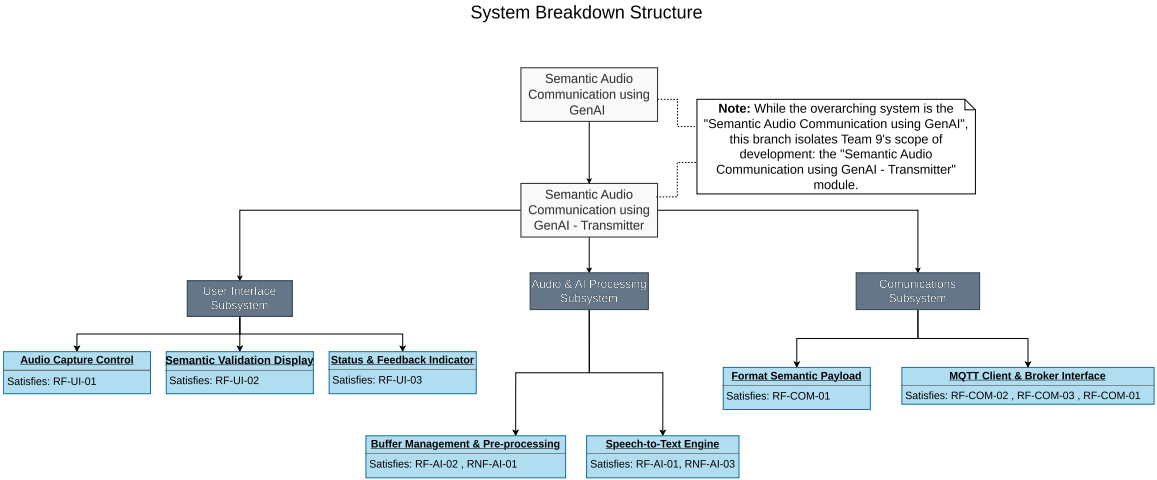

**AI PROCESSING: Whisper large-v3**

- High Accuracy STT & Native Portuguese Support.
- 100% Offline Capable (Critical for live demos)
- Free & Open-Source

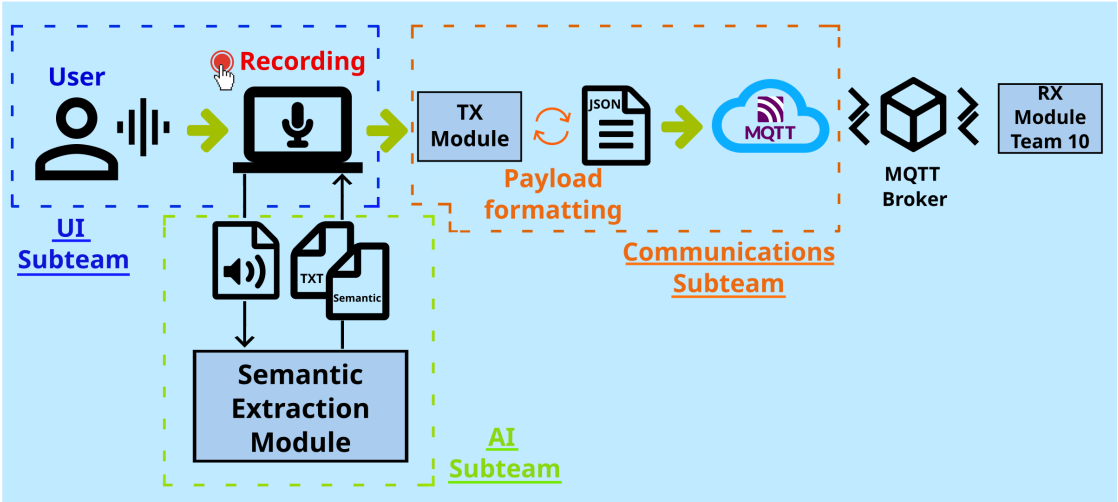
**COMMUNICATION TRANSPORT: MQTT**

- Minimal network overhead & very low latency
- Pub/Sub Broker Paradigm
- Asynchronous delivery (Decouples TX from RX)

# System Breakdown Structure



# Functional Architecture



# Measures of Effectiveness and Performance

## WER

### Semantic Accuracy

- Evaluates the system's effectiveness in capturing the users's correct intent.
- If the WER is high, the main objective semantic communication) fails.

## Delivery Reliability

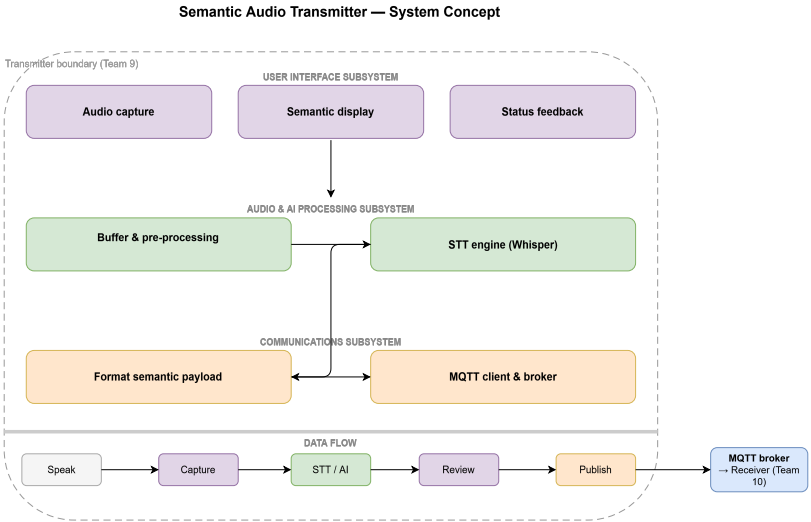
- The percentage of payload that are successfully published to the Broker out of the total number of attempts.
- The percentage of payloads that are compliant with the schema chosen to format the payload (JSON).

## End-to-End Latency

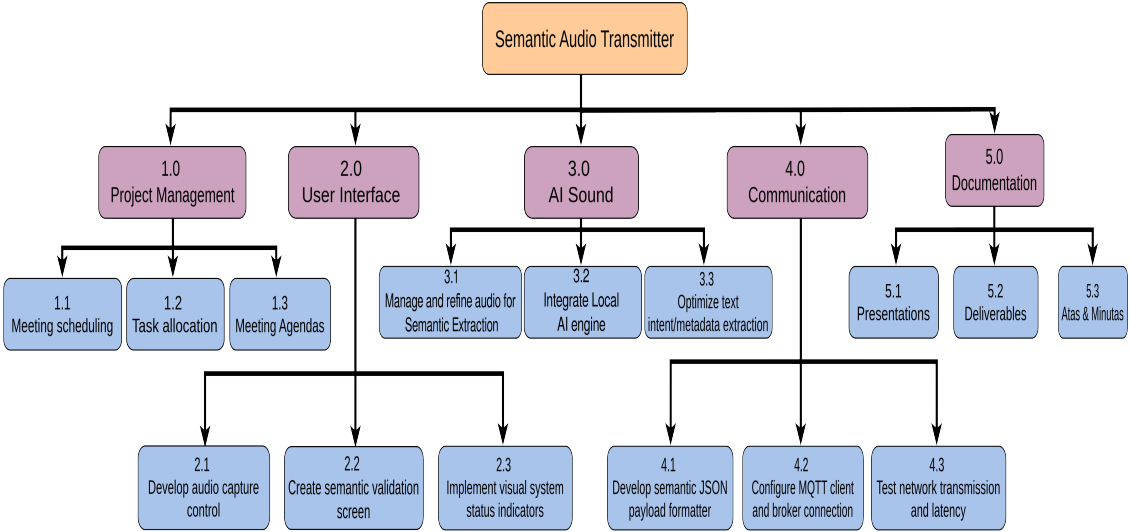
### [MQTT Delivery Reliability]

- The total time measured from the moment the user finishes speaking until the formatted payload is sent.
- The risk report stipulates that the model's latency cannot exceed the defined limit.

# System Concept



# Work Breakdown Structure + Gantt

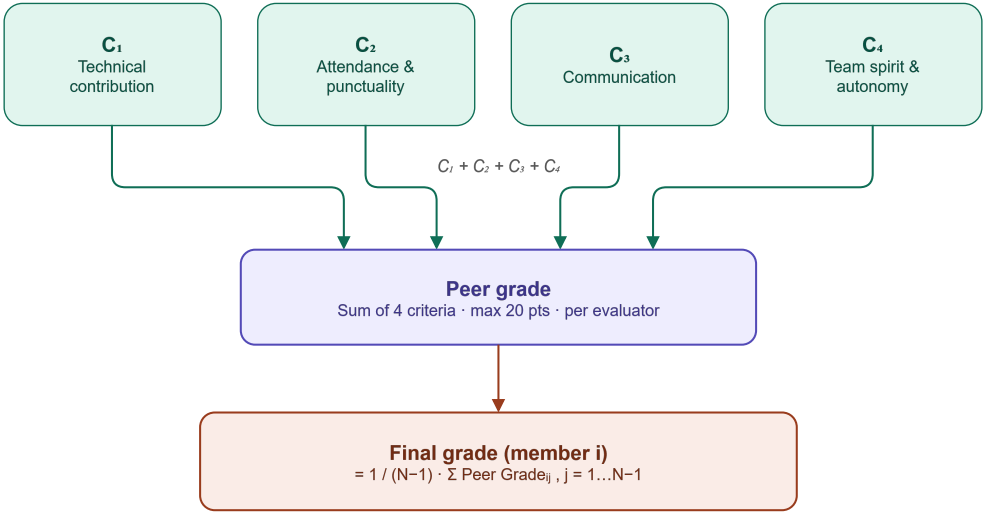


# Risk Management

| PoO \ Impact | Low | Medium | High |
|---|---|---|---|
| **High** | | | High Latency From AI Model |
| **Medium** | | Insufficient Audio Quality | MQTT Communications Failures |
| **Low** | | Resource Exhaustion During Demos | HW Failures |

**PoO** - Probability of Occurence

# Internal Evaluation



# End of Semester Plan

- The next step is to move from preliminary design into concrete implementation of the user interface, the audio-conditioning chain, the semantic processing module, the payload builder, and the MQTT client [8].
- In parallel, the team needs to consolidate the transmitter-receiver interface, keep architectural consistency with previous deliverables, and continue integration testing under realistic local-demo conditions [9, 8].
- The end-of-semester objective is a stable demonstrator that preserves the validated requirements, supports local operation, and is ready for evaluation and live presentation [5, 7, 8].

# References I


[1] *Guide to the Systems Engineering Body of Knowledge.*
INCOSE, 2.13 edition, 2025.

[2] adbista.
mqtt-voice-chat: A gui-based mqtt communication application.
https://github.com/adbista/mqtt-voice-chat.
Accessed: 2026-04-21.

[3] MattoMatteo.
Homegpt: Mqtt voice assistant.
https://github.com/MattoMatteo/HomeGPT.
Accessed: 2026-04-21.

[4] pinootto.
voice-mqtt: A voice interface for mqtt.
https://github.com/pinootto/voice-mqtt.
Accessed: 2026-04-21.

# References II


[5] Team 9.
Deliverable 1: Analysis and requirements specification — semantic audio communication using genai – transmitter.
Technical Report Deliverable 1, Faculty of Engineering, University of Porto, March 2026.

[6] Team 9.
Deliverable 3: Market survey — semantic audio communication using genai – transmitter.
Technical Report Deliverable 3, Faculty of Engineering, University of Porto, March 2026.

[7] Team 9.
Deliverable 5: System concept — semantic audio communication using genai – transmitter.
Technical Report Deliverable 5, Faculty of Engineering, University of Porto, March 2026.

[8] Team 9.
Deliverable 6: Preliminary design — semantic audio communication using genai – transmitter.
Technical Report Deliverable 6, Faculty of Engineering, University of Porto, April 2026.

# References III

[9] Team 9.
*Systems Engineering Team Manual: Semantic Audio Communication using GenAI – Transmitter.*
Faculty of Engineering, University of Porto, March 2026.